2.Chebyshev IIR Filters

2.1 Design and performance Evaluation of a Chebyshev Type IIIR Low Pass Filter using MATLAB

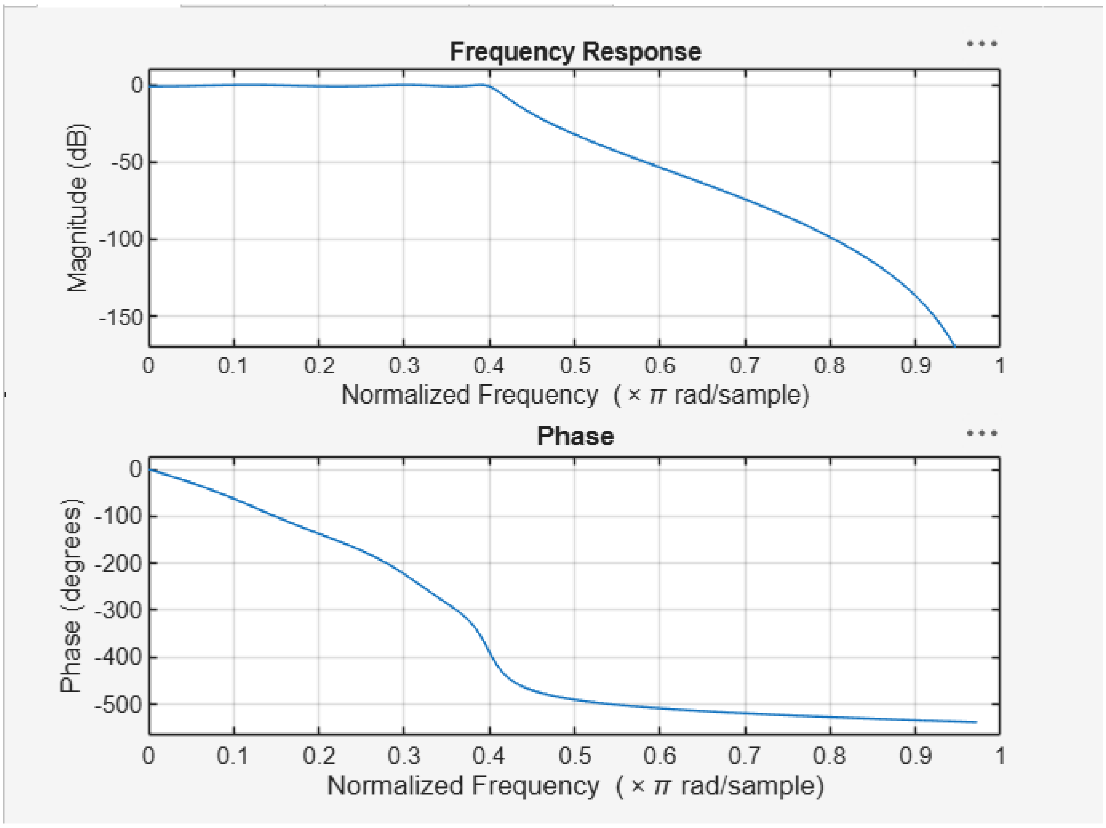
PROGRAM:

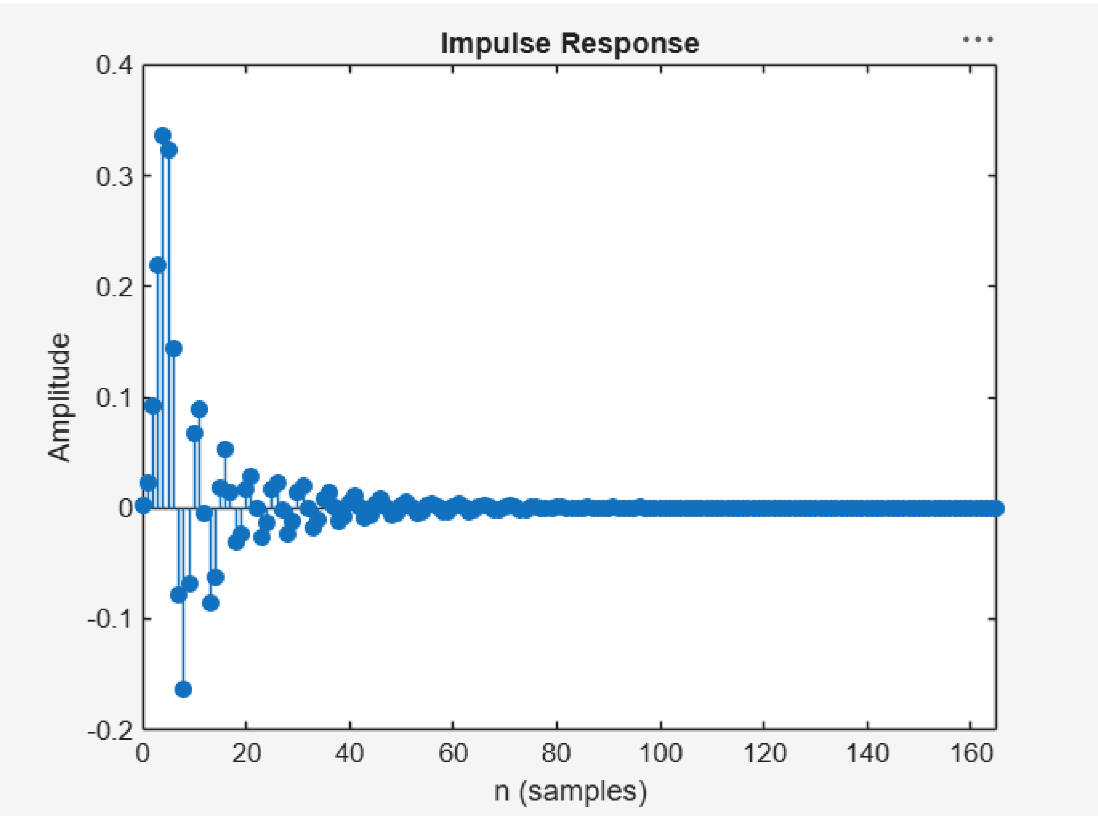
```
clc;

clear;

close all;


% Specifications

Wp = 0.4;      % Passband edge frequency

Ws = 0.55;     % Stopband edge frequency

Rp = 1;        % Passband ripple (dB)

As = 40;       % Stopband attenuation (dB)


% Find filter order and cutoff frequency

[N, Wn] = cheb1ord(Wp, Ws, Rp, As);


% Design Chebyshev Type-I Low Pass Filter

[b, a] = cheby1(N, Rp, Wn);


% Display results

fprintf('Filter Order = %d\n', N);

fprintf('Cutoff Frequency = %.4f * pi rad/sample\n', Wn);


% Stability Check
```

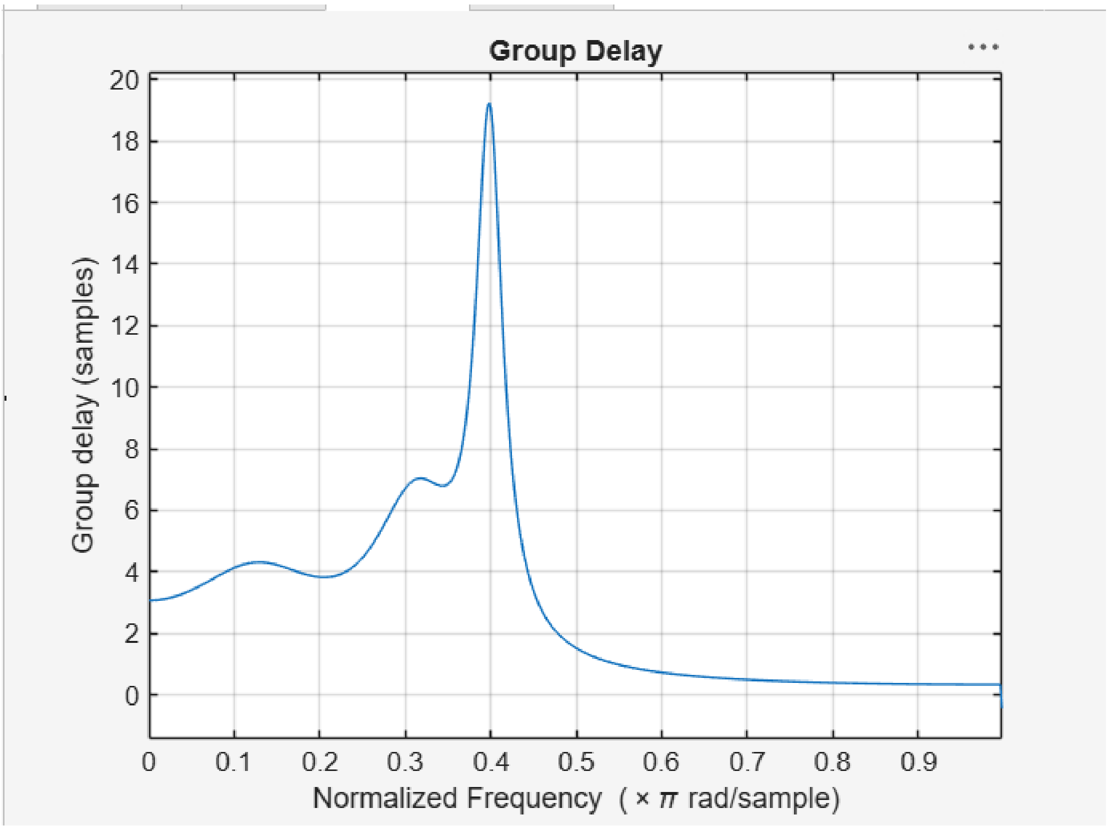
```
if all(abs(roots(a)) < 1)
    disp('System is STABLE');
else
    disp('System is UNSTABLE');
end

% Magnitude and Phase Response
figure;
freqz(b, a);
title('Frequency Response');

% Impulse Response
figure;
impz(b, a);
title('Impulse Response');

% Group Delay
figure;
grpdelay(b, a);
title('Group Delay');

% Pole-Zero Plot
figure;
zplane(b, a);
```

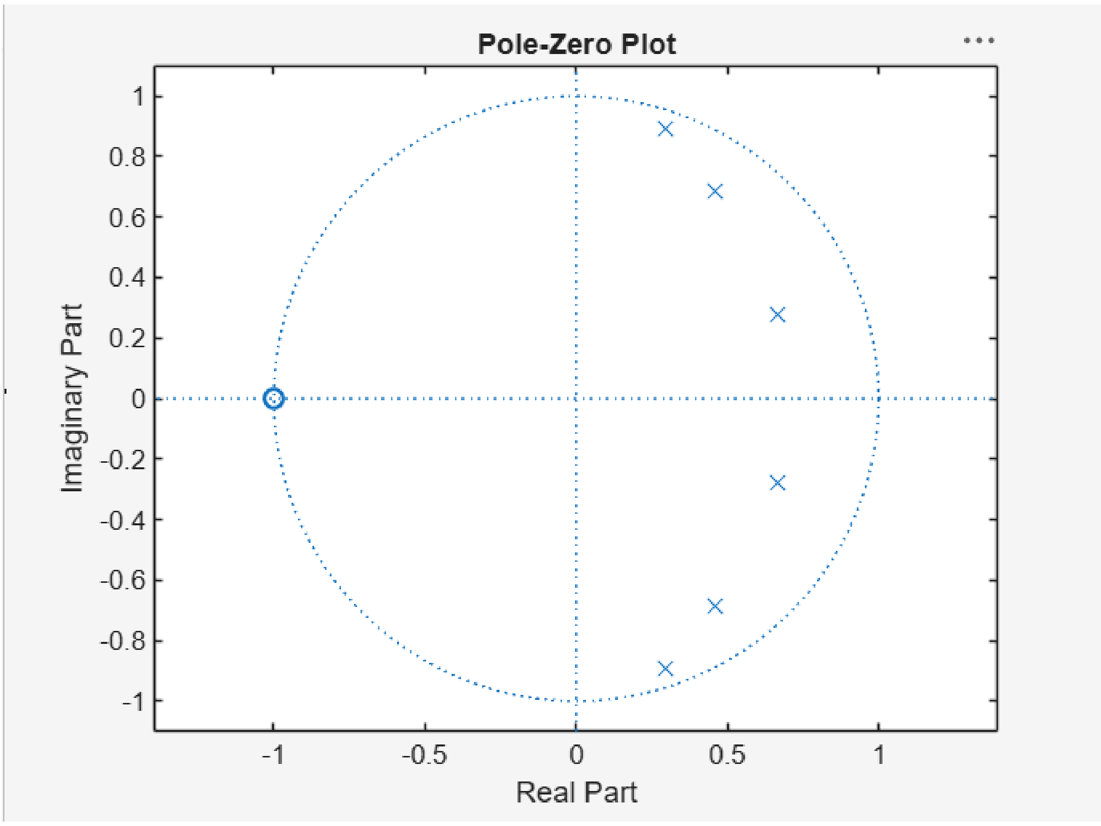
```
title('Pole-Zero Plot');
```

OUT PUT:


Frequency Response
Magnitude (dB)
0
-50
-100
-150
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
0
-100
-200
-300
-400
-500
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1
Normalized Frequency (× π rad/sample)


Impulse Response
Amplitude
0.4
0.3
0.2
0.1
0
-0.1
-0.2
0 20 40 60 80 100 120 140 160
n (samples)






```
Cutoff Frequency = 0.4000 * pi rad/sample
System is STABLE
>>
```

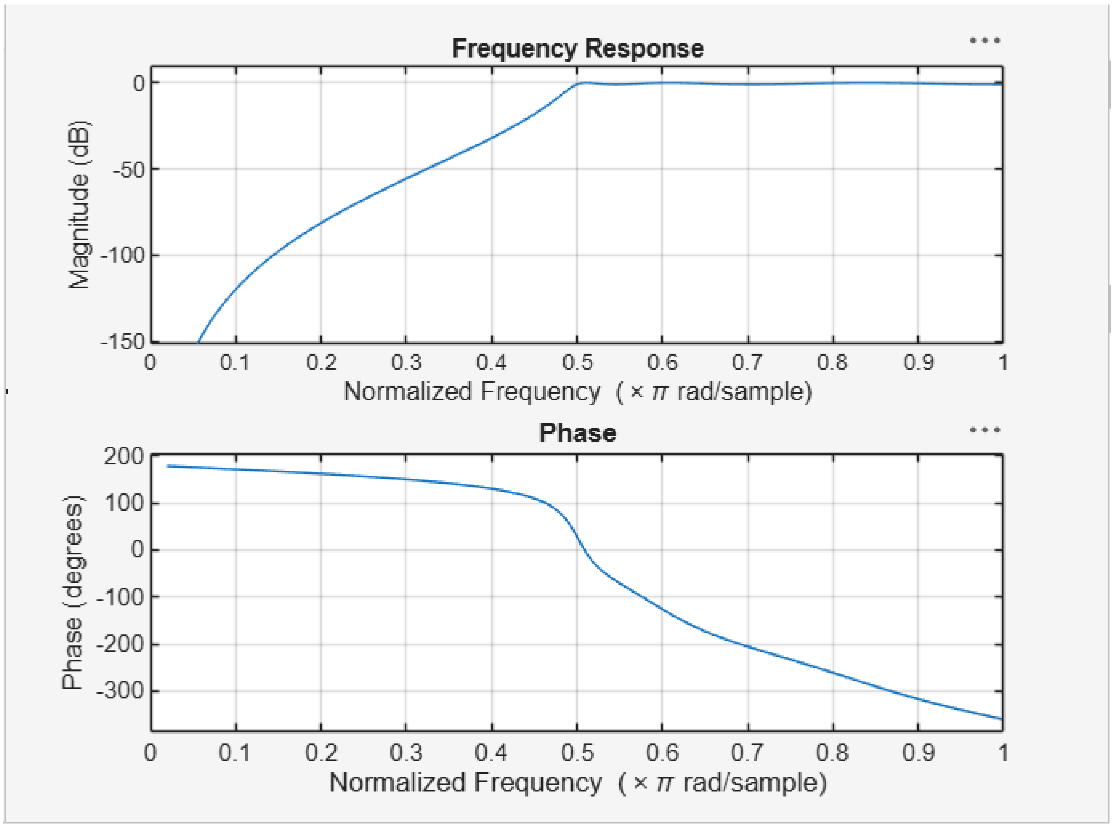
2.2 Design and performance Evaluation of a Chebyshev Type IIIR High Pass Filter using MATLAB

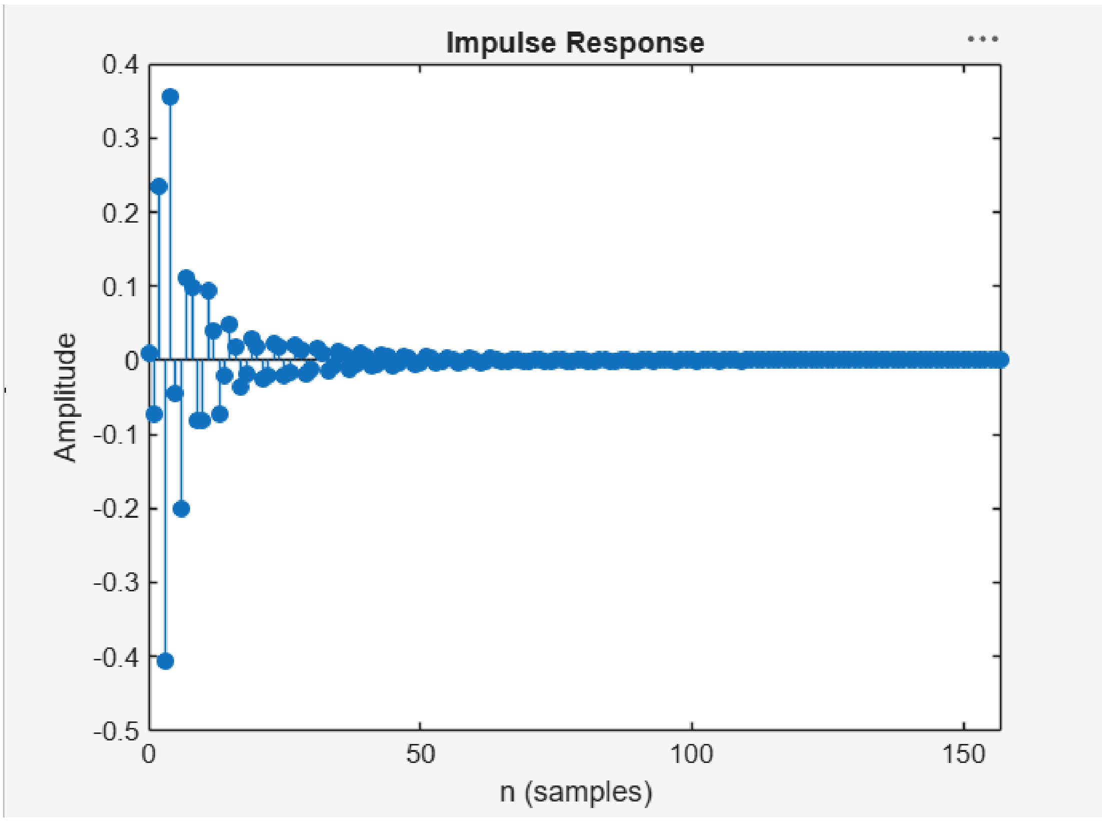
PROGRAM:

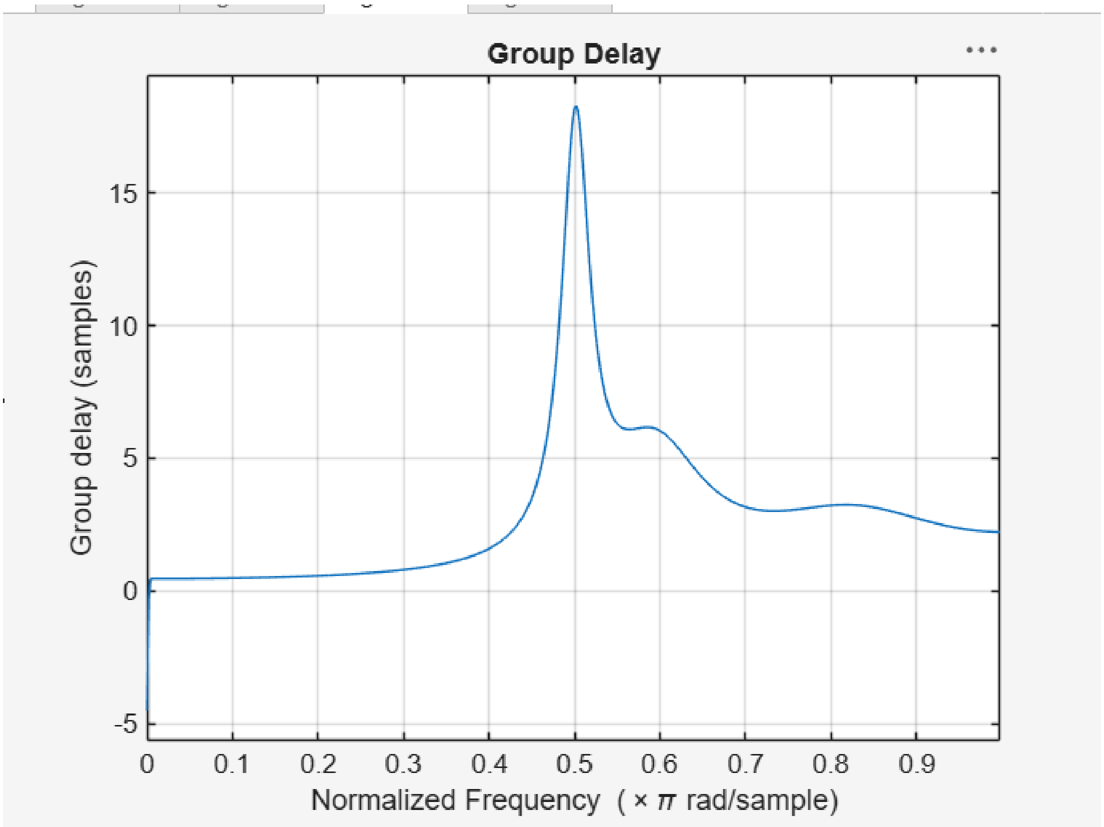
```
clc;

clear;

close all;

% Specifications

Rp = 1; % Passband Ripple (dB)

Rs = 40; % Stopband Attenuation (dB)

Wp = 0.5; % Passband Edge

Ws = 0.35; % Stopband Edge

% Find Filter Order

[n,Wn] = cheb1ord(Wp,Ws,Rp,Rs);

% Design High-Pass Filter

[b,a] = cheby1(n,Rp,Wn,'high');

% Frequency Response

figure;

freqz(b,a);

title('Frequency Response');

% Impulse Response

figure;

impz(b,a);

title('Impulse Response');

% Group Delay
```

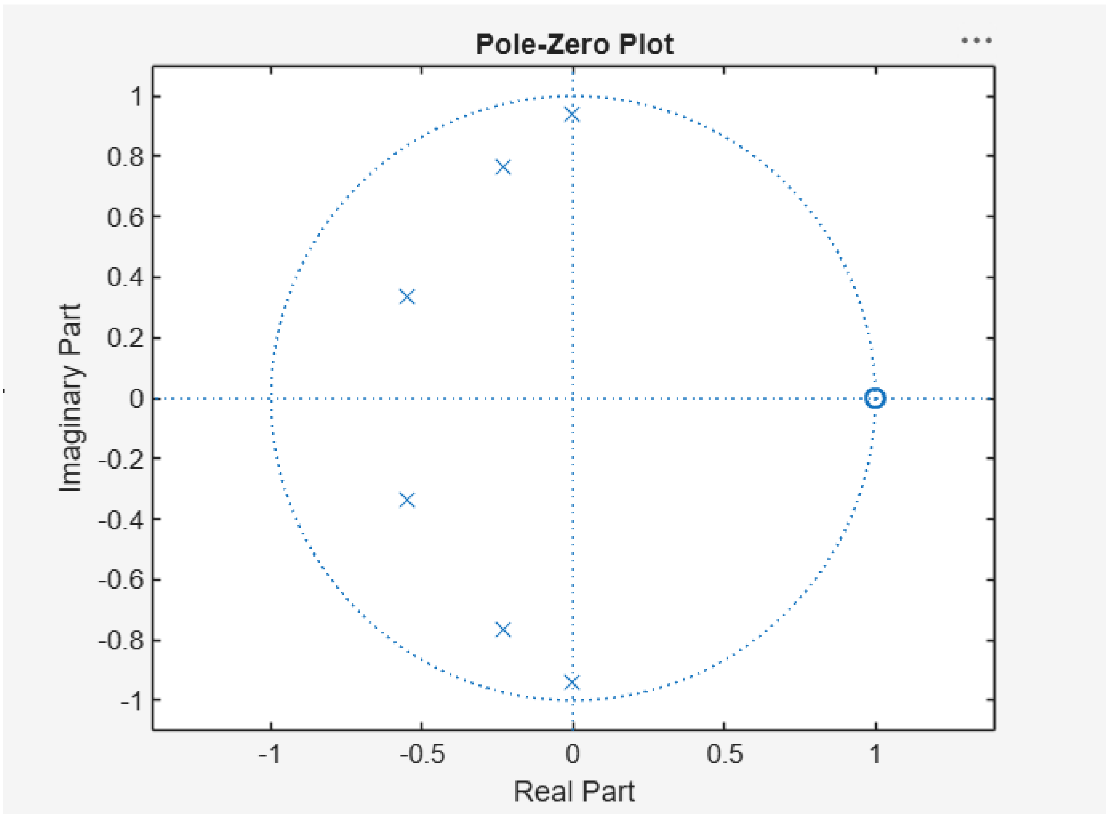
```
figure;

grpdelay(b,a);

title('Group Delay');

% Pole-Zero Plot

figure;

zplane(b,a);

title('Pole-Zero Plot');

fprintf('Filter Order = %d\n',n);

fprintf('Cutoff Frequency = %.4f\n',Wn);
```

OUT PUT:


Frequency Response
Magnitude (dB)
0
-50
-100
-150
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
200
100
0
-100
-200
-300
0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1
Normalized Frequency (× π rad/sample)


Impulse Response
Amplitude
0.4
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
-0.4
-0.5
0 50 100 150
n (samples)


Group Delay
Group delay (samples)
15
10
5
0
-5
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
Normalized Frequency (× π rad/sample)


Pole-Zero Plot
Imaginary Part
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1
-0.5
0
0.5
1
Real Part

```
Filter Order = 6
Cutoff Frequency = 0.5000
>>
```